

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER EXAMINATION

SUMMER SEMESTER 2016-2017

FULL MARKS: 75

DURATION: 1 Hour 30 Minutes

CSE 4645: Numerical Analysis

Programmable calculators are not allowed. Do not write anything on the question paper.
There are 4 (four) questions. Answer any 3 (three) of them.
Figures in the right margin indicate marks.

- 8 While solving a mathematical model using numerical methods, how can we use relative approximate errors to minimize the error? If anyone wants at least m significant digits to be correct in the answer, then what you would need to do? 5
- 9 Answer the followings about Taylor series: 5+5
 - i. Why are the applications of Taylor's theorem important for numerical methods?
 - ii. Why the Taylor's Polynomial cannot be used for interpolation?
- 8 The Taylor series for e^x at point $x = 0$ is given by 2+8

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \quad -\infty < x < \infty$$
 - i. What is the truncation (true) error in the representation of $e^{2.5}$ if only four terms of the series are used?
 - ii. How many terms it would require to get an approximation of $e^{2.5}$ within a magnitude of true error of less than 10^{-6} ?
- 5 a) Use the graphical approach to determine the drag coefficient c needed for a parachutist of mass $m = 68.1$ kg to have a velocity of 40 m/s after free-falling for time $t = 10$ s. The acceleration due to gravity is 9.8 m/s². Use False position method to obtain your solution. Use at least two iterations to obtain the root. 9
- b) Use the Newton-Raphson method to estimate the root of $f(x) = e^{-x} - x$, employing an initial guess of $x_0 = 0$. Employ an initial guess of 0 and perform at least two iterations. 8
- c) What are advantages and disadvantages of Bisection method for root finding problem? Briefly explain. 8
- 1 a) Use Müller's method to determine the positive real root of $f(x) = x^3 + x^2 - 4x - 4$. Perform at least two iterations to find out the approximate root. 15
- b) Show that Fixed Point theorem does not ensure a unique fixed point of $g(x) = 3^{-x}$ on the interval $[0, 1]$, even though a unique fixed point on this interval does exist. Explain your answer and determine how we can fix the problem to approximate to this fixed point to any specified degree of accuracy. 5
- c) What is Inflection Point? Explain. 9
- 4 a) Use Neville's method to approximate $\sqrt{5}$ using two degree approximation with the following functions and values. $f(x) = e^x + \sqrt{x}$ and the values $x = 0, 1, 2, 3, 4$ 8
- b) Why minimize the sum of the square of the residuals in case of regression? Briefly explain. 4
- c) When n becomes large, in many cases, one may get oscillatory behavior in the resulting polynomial interpolation. This was shown by Runge when he interpolated data based on a simple function of $y = \frac{1}{1+25x^2}$ on an interval of $[-1, 1]$. So what is the solution if you want to use information from more data points, but at the same time keeping the function true to the data behavior? Explain your answer why your approach will lead to better solutions. 4
- d) How does regression differ from interpolation? Explain.